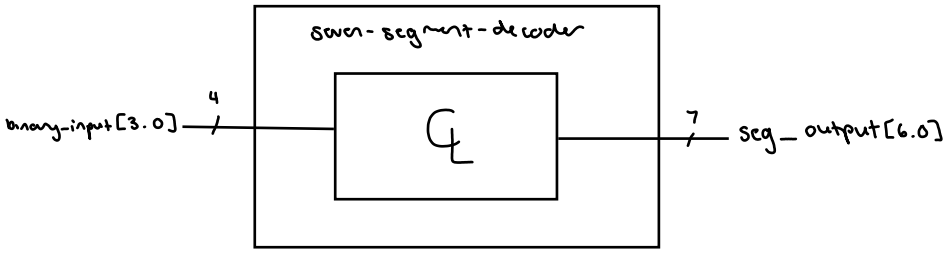
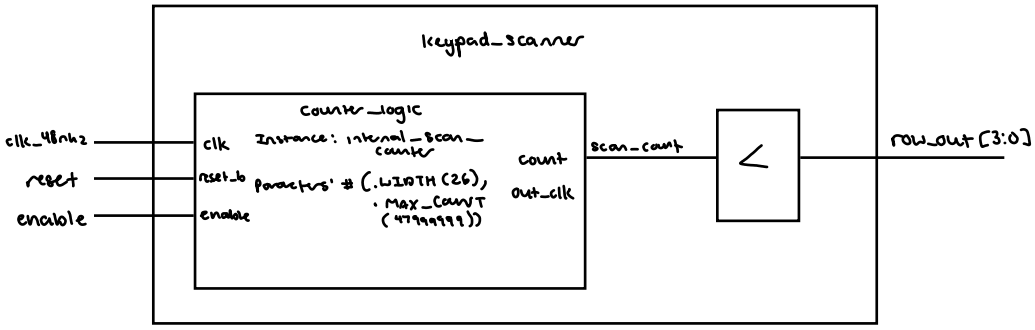
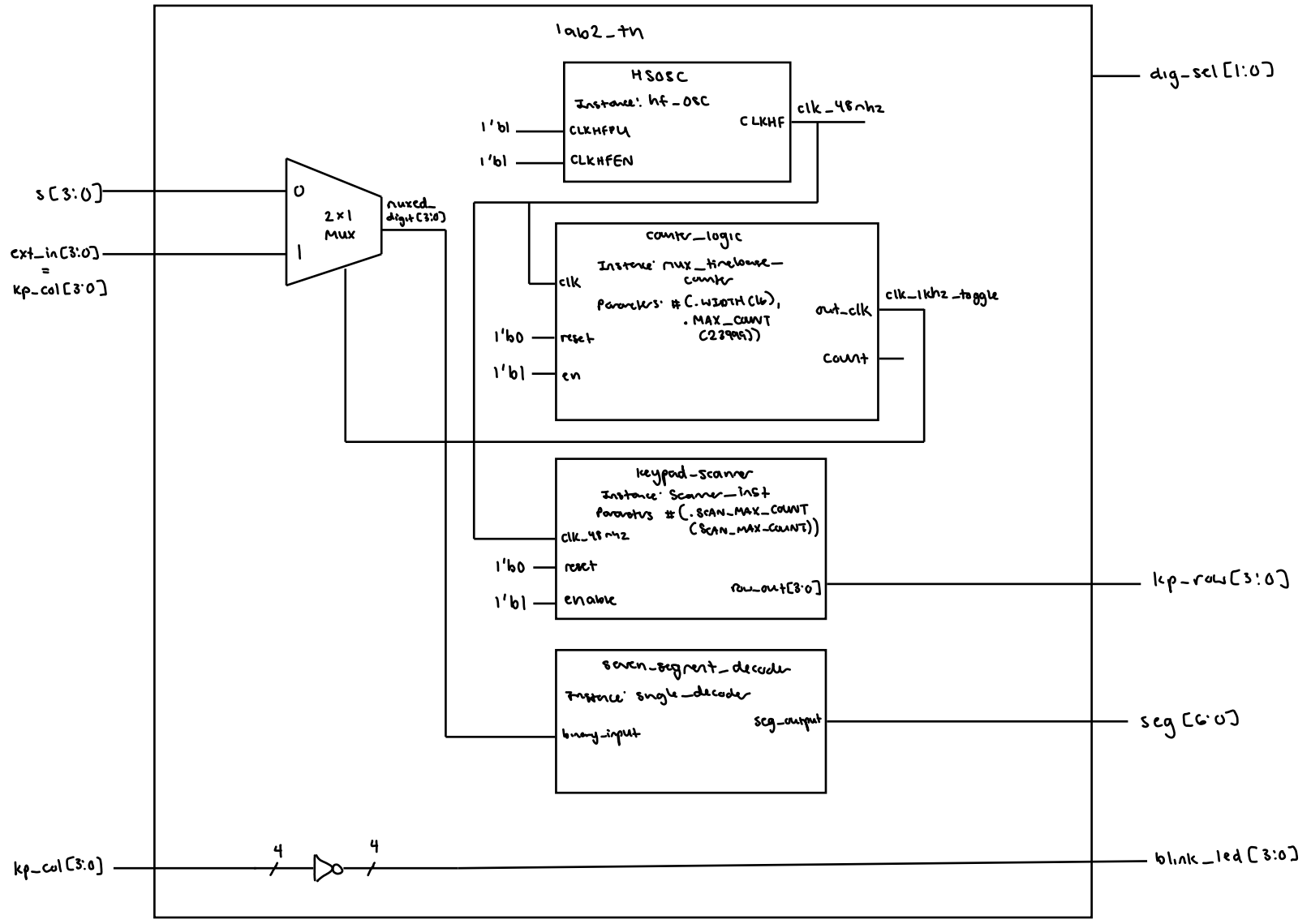

counter_logic
clk
enable
reset_b
count [23:0]
out_clk
MAX_COUNT


seven-segment-decoder
binary_input[3.0]
4
7
seg_output[6.0]


keypad_scanner
counter_logic
clk_48mhz
reset
enable
clk
reset_b
enable
Instance: internal_scan_counter
parameters #(.WIDTH(26), .MAX_COUNT(47999999))
count
out_clk
scan_count
row_out[3:0]


lab2_tn
HSOSC
Instance: hf_OSC
1'b1
CLKHFPU
1'b1
CLKHFEN
CLKHF
clk_48mhz
dig_sel[1:0]
s[3:0]
0
2x1
Mux
1
muxed_
digit[3:0]
ext_in[3:0]
=
kp_col[3:0]
counter_logic
Instance: mux_timebase_
counter
clk
Parameters: #(.WIDTH(16),
.MAX_COUNT
(23999))
out_clk
clk_1khz_toggle
1'b0
reset
count
1'b1
en
keypad_scanner
Instance: Scanner_inst
Parameters #(.SCAN_MAX_COUNT
(SCAN_MAX_COUNT))
clk_48mhz
1'b0
reset
row_out[3:0]
kp_row[3:0]
1'b1
enable
seven_segment_decoder
Instance: single_decoder
seg_output
seg[6:0]
binary_input
kp_col[3:0]
4
4
blink_led[3:0]